

From the Master's Tool to an Instrument of Change: HCI Design Method Subversion

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Abstract

The dominant view of HCI design methods, aligned with profit-driven goals, reinforces existing power structures and economic inequality. Inspired by Audre Lorde's assertion that "the master's tools cannot dismantle the master's house," this paper examines their potential to enable socially just, community-driven outcomes by subverting conventional methods like example mapping. We apply Amos Vogel's framework of subversion of form and content, using case study in cooperative and community land trust context. This research represents a novel approach as it specifically focuses on subverting a conventional design method to align with cooperative principles and economic justice. Findings highlight the importance of inclusivity, mutual aid, and community-centred processes in shifting technology design goals toward equity and collective ownership.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **HCI design and evaluation methods**;

Keywords

design methods subversion, example mapping, design sprint, participatory, design, critical design

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1 INTRODUCTION

Current extreme economic inequality and the intense concentration of wealth are perpetuated, in part, by the design of interactive systems and digital infrastructure that prioritize profit maximization for the few over equitable access and opportunity for the many [32, 38, 50, 55]. We start with an assumption that industry-standard

design paradigms for such systems and infrastructures often reinforce existing power structures by embedding biases that favour corporate interests, algorithmic surveillance and extractive data practices [64, 68]. To address this, as reimagining design methods in human-computer interaction (HCI) is necessary, we propose an approach inspired by Audre Lorde's assertion that "the master's tools cannot dismantle the master's house" [56]. Subverting a UX design method, example mapping, to address the problem as expressed by Lorde, we explore how prioritizing community-driven, participatory design processes that explicitly include co-operative principles may challenge entrenched hierarchies and build interactive systems rooted in economic justice. We explore the early-stage design process for community-owned technology, focusing on small and medium-sized co-operatives and community land trusts (CLTs). Grounded in feminist scholarship, we are motivated in this work by the research question: How participatory and co-design methods for developing interactive systems and digital infrastructure can support community wealth building?

To contextualize the approach to our exploration, we look at some of the broader conditions that influence the conventional use of design methods in this space. Computer science, and later HCI, has since the Second World War been driven by a 'military-industrial' complex [27, 52, 61] with an underlying aim of technological, economic and political dominance. The military-industrial complex, originally coined by U.S. President Dwight D. Eisenhower in 1961, refers to the relationship between a country's military, its government and industries that produce defense equipment and technology. In this context, digital technology, over the last thirty years, has evolved to encompass broader Silicon Valley's ideological influences [3]. These merged the military-driven objectives of speed and stealth with an entrepreneurial and libertarian ethos. This hybridization has played a defining role in the way HCI design reinforces the status quo by aligning technological innovation with market-driven values and practices [32, 50]. A more critically orientated HCI recognizes and critiques these goals and objectives as incompatible with an ethically driven view of technology [20, 26, 27, 34, 40, 49, 62]. These critical investigations make space for new design methods, or subversion of design methods, through which technology becomes an instrument of social change rather than a tool of control [41]. Furthermore, the subversion of the design method we present here is part of our commitment to HCI research that counters "how (technological) initiatives become yet further a means of exploiting for profit" [55]. We draw on the ways in which subversion in HCI is productively considered a tool [4], a strategy [2], a theoretical lens [43] and a form of resistance

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[53, 63]. Drawing on and moving beyond contemporary debates in this space, this paper adds to the social justice, feminist, decolonizing and ethics focused literature in HCI by developing a critical exploration of how we might subvert a design method.

In this paper we outline how we modified this design method in order to subvert the original goals of efficiency and productivity, and instead forward the principles of mutual aid, cooperation and feminist goals. Through the work presented in this paper, we outline initial findings from our case study, which offer a novel way for HCI researchers and designers to engage with communities and the emancipatory work that they do through shared understanding of community goals [10]. We discuss findings using Amos Vogel's classification "subversion of form" and "subversion of content" [67] to develop a systematic approach to shaping technology that helps dismantle the 'master's house' [56]. We conclude the paper by proposing that the example mapping subversion may benefit community wealth building and heed a call to more research in economic justice-focused HCI.

2 BACKGROUND AND RELATED WORK

2.1 Example mapping and its limitations

Example mapping is an under-explored collaborative UX design method developed as part of a software development approach that helps design teams understand how software should behave [69]. In example mapping sessions key representatives from three perspectives—business, development, and testing—come together to define user stories, rules and examples in a visual or written format [10]. However, this method reveals inherent limitations when viewed through a critical lens. While it facilitates conversation, it privileges voices already skilled and experienced in design language and decision-making processes. This reinforces the status quo rather than challenging it, echoing Audre Lorde's assertion that "the master's tools will never dismantle the master's house." By focusing on granular details, example mapping aids perpetuating a narrow, technocratic approach to problem-solving, one that overlooks the broader socioeconomic, environmental and political dynamics at play. As such, it replicates the problematic role of design, rather than contributing to positive transformative change.

2.2 Participatory design

Whilst historically set to improve working conditions and democratizing the workplace, in the context of economic inequality, PD emerges as a crucial approach, as it seeks to involve marginalized communities in the decision-making processes that affect their economic realities [11]. Pioneering scholars like Suzanne Bødker have laid the groundwork for critical PD research [11, 13–16]. In particular we draw on the call to "re-emphasize the foundations of [PD] in the service of enabling deeper local impact, based on mundane, everyday, and socio-material and political engagements" [12]. We are interested in PD as democratic inquiry as DiSalvo expresses it [28], advocating for public good over corporate interests and critiquing innovation's role in reinforcing inequality. Additionally, we draw on Haskel's research on open-source and co-operative design [45] challenges traditional dominant structures, promoting inclusive outcomes and community autonomy. Moreover, we are

encouraged by Harrington's and D'Ignazio's exploration of equitable PD as a form of design activism [23, 44]. Lastly, expanding beyond traditional PD and HCI literature into media and cultural studies, we draw on Barney et al. [7] who see participatory culture in digital age a necessity to resist corporate influence and promote genuinely democratic practices. We build on this work to highlight PD's potential for subversion—transforming 'master's tools' into instruments of bottom up change.

2.3 Feminist HCI

Feminist HCI is an evolving area within HCI that integrates feminist theories and practices to promote inclusion, accountability, and social justice in design. It contributes essential insights into the role of technology in shaping social and economic relationships; this includes the need to consider how pluralist approach and advocacy can help open the design process to broader participation [5]. For example, Bardzell [5, 6] highlights the importance of feminist commitments such as equity, diversity, and social justice, throughout design. Breslin & Wadhwa [18] and Bellini et al. [9] advocate for feminist engagement and community-building in HCI, while Kumar et al. [51] emphasize feminist solidarity in resource-constrained regions. Empirical studies have applied these principles, such as feminist participatory design in breast pumps, crossing over to CSCW [24, 25], value-driven design in fan fiction archives [36], gendered experiences in IT [1], and sexual consent in HCI [59]. Recent work by Croon and Key et al. [22, 48] explores care as a design strategy and challenges androcentric approaches, aligning HCI with feminist critiques of universality, objectivity, tacit knowledge, and lived experiences. Research conducted within a feminist framework is attentive to difference, social power, and a commitment to activism and social change [47], providing a full set of tools to interrogate HCI processes. Our research builds on these foundations, focusing on marginality, plurality, hybridity of models and methods, and the principles of "resist and build"—serving as a reflection on and reminder of its feminist purpose.

2.4 Subversion in HCI

Subversion in HCI is an emerging concept. The term subversion generally refers to an act of undermining, overturning, or challenging established systems, structures, or norms. It involves disrupting or altering something—often in a way that challenges authority, conventions, or the status quo. In HCI scholarship, subversion is mostly explored as a methodological strategy challenging normative practices in design. For example, subversion is seen as a form of design action that acknowledges the political nature of design, troubling conventional technology design practices that often ignore or exacerbate environmental degradation [55]. It is also employed as a method for enhancing civic engagement, challenging the status quo by proposing alternative design practices that question corporate interests, consumerism, and the centralization of power in design decision-making [49]. Additionally, subversion functions as a participatory design process, ensuring that the design process is not only about producing usable technology but also about empowering communities, giving them the tools to address their own issues and concerns through co-creation [21]. Finally, subversion as creative action works to rethink design processes by introducing

ethical, feminist, and social justice frameworks into mainstream HCI discourse, furthering subversive participatory design within the HCI social justice framework and emphasizing its potential to challenge and disrupt traditional power dynamics [2]. This body of work highlights the importance of active, critical methodological engagement we wish to expand to a wider range of cross-disciplinary studies in HCI.

2.5 Community wealth building and feminist economics

Community wealth building addresses economic inequality by shifting economic power and resources to local, community-based organizations and individuals, emphasizing collective ownership, democratic governance, and equitable wealth distribution to create more resilient, sustainable economies that provide opportunities for marginalized groups [31, 57, 70]. Researchers who have looked at this subject argue that HCI can play significant role in supporting such work in form of solidarity economy [8, 65, 66] as an umbrella framework within which community wealth building occurs. Moreover, Chordia et al. [19] and Fox et al. [37] invite the HCI community to develop concrete strategies for a design that address economic inequality as a part of a broader commitment to social justice. Furthermore, in doing the work on subversion aiming at economic justice, we have drawn inspiration from scholarship outside of HCI, including the feminist economy. Feminist economic research offers a complementary exploration of subversion through the lens that helps rethink traditional economic structures, including our understanding of community wealth. The seminal work of Gibson-Graham [42] proposes alternative frameworks, such as difference and economic plurality, that critique the dominance of a monolithic approach to the economy. Similarly, expanding this discourse, feminist subversion of the economy [58] offers additional evidence that this approach is a helpful way of addressing inequality.

Building on this foundation the following section introduces process through which we subvert example mapping method as a focal point for our exploration. By applying this novel approach, we aim to explore how conventional design methods can be transformed to challenge dominant paradigms and better serve the interests of people experiencing economic inequality.

3 STUDY

The study involved a series of four participatory sessions held in London between October 2023 and September 2024. The overarching goal of these workshops was to explore and reflect on the use of co-operative principles as a means to subvert the example mapping method. The sequential progression of the sessions provided us with the continuity needed for iterative learning, which helped us articulate the purpose and form of the subversion. In particular, the first stage established a foundation for understanding the co-operative design space. Building on this, the second stage introduced 'solidarity mapping' as a subversive approach. Finally, the last stage focused on community land trust members testing and evaluating modifications to the method, providing a specific use context.

The participants in our workshops represented a diverse range of co-operative members and the general public, including workers in digital co-operatives, housing co-operatives, and community land trusts. Some participants were new to this space, while others brought decades of experience in cooperative organizing and economic justice efforts dating back to the 1980s, creating a rich foundation for participation. What follows is a short description of these sessions highlighting key examples, reflections and preliminary insights:

3.1 Co-op Hackathon: Introducing example mapping

3.1.1 Aims and context summary: The first session, part of the first ever UK Co-op Hackathon [71], marked the beginning of a series of workshops for this study, aimed at exploring and reflecting on the use of co-operative principles within example mapping. Ten participants who engaged in a 1-hour workshop ranged from experienced co-op members to beginners. The central activity focused on example mapping, where participants were introduced to the method's history and previous use in an art project on personal data and privacy [72]. Participants explored printed materials on co-op principles and engaged in discussions around four prompts central to the method: "rule", "story", "example", and "questions", focusing on ongoing projects and grassroots co-design challenges. The activity produced five detailed plates with narratives on various themes, such as co-operative art education, community ownership of unregistered land, platform co-operatives, blockchain-based decentralization, and collaborative funding for community solar power.

3.1.2 Key examples:

- **Solar energy funding:** Emphasized community-driven values, and ethical financial practices.
- **Blockchain finance:** Highlighted the potential of decentralized finance to enhance co-operatives' financial autonomy and operational efficiency.
- **Unregistered land use:** Focused on using technology to empower residents to address complex issues related to community wealth building through asset acquisition.

3.1.3 Reflection and iteration: The examples and questions encouraged our reflection on potential challenges, creating a cyclical process informing future conversations and stages of design. Most importantly, the example mapping at this level highlighted key aspects of community wealth building, such as community ownership, democratic governance, building local capacity, and the equitable distribution of wealth. It also revealed problems and tensions related to the use of digital infrastructure and interactive systems in this space that, by design, fail to support or include any of co-operative principles or goals. These insights suggested participants' interest in reframing design methods toward emancipatory goals, such as creating digital infrastructures and interactive systems built on co-operative principles. This helped us develop better understanding of the design space and co-operative technology community perspectives and laid the groundwork for the next stage of research.

3.2 The Sandbox sessions: Looking for subversion

3.2.1 Aims and context summary. Building on the Co-op Hackathon experience, we designed the Sandbox sessions to explore design methods through collective scrutiny, focusing on identifying opportunities for *subversion*. The two 2-hour workshops included a presentation, a Q&A session, and reflective exercises where participants considered various existing methods, including example mapping, to discover ways to modify them to better meet the community's needs. Two-thirds of the participants had also attended the Co-op Hackathon session, providing continuity and shared context for the discussions. Participants engaged in conversations to uncover and discuss key elements, including "rules" (policy barriers), "stories" (local examples of economic challenges), and "questions" (how technology or grassroots efforts might address these challenges). Two sessions generated data that captured participants' thoughts on topics such as community-led design and representation, skill accessibility and development, multilayered economic participation, balancing technical and community insights. We collected responses to questions about co-creating software within community groups, the role of local economies in community wealth building, and the main challenges and opportunities of community co-created technology in collaborative funding decisions.

3.2.2 Key examples:

1. Participants questioned who decides to allocate community funds and emphasized the need for a system that reflects community values.
2. Participants highlighted the need to provide groups without design, computer science or HCI training with menus of possibilities and opportunities to engage in interactive system co-design process.
3. Participants highlighted the need to create communication conduits for those who do not have written or spoken English.

3.2.3 Reflection and iteration: Reflection and iteration at this middle stage of our study involved analysing different ways to structure and organize community collaboration, potentially subverting traditional hierarchical models and challenging conventional economic structures' contributions to community wealth building. This process importantly led us to consider focusing on "solidarity mapping" as a way to subvert both the form and content of the example mapping design method. Moreover, based on the conversations and data collected in this stage of research, we were reminded of and influenced to focus on community-based practices of care through informal savings and other forms of asset building, such as community land trusts.

3.3 Community land trust away day: Solidarity mapping

3.3.1 Aims and context summary: The last stage of our research into subversion of example mapping involved a community land trust, LDN.CASH (Community Assets for Society and Housing) based in South-East London [73]. Having transitioned from general explorations of co-operative design as a point of entry for

collaboratively imagining and envisioning subversion, we were now prepared to engage a specific community group. In preparation for this cumulative stage, we considered how to explore and define the key aspects of their initiatives, ensuring that every decision prioritized equity, solidarity, and collective benefit. This approach aimed to advance and deepen the collaborative aspects of the research process by allowing participants to modify methods, grounding the process in equity and feminist principles—not only for the outcomes but also for the process itself. During the away day session and subsequent informal feedback the modified mapping tool was noted as valuable for both new and existing co-ops, aiding in future development planning. Customization of form to fit different visions and scenarios was deemed essential. The 'story' aspect was praised for its versatility, while establishing guiding principles instead of 'rules' was seen as crucial for setting up appropriate tone for conversations leading towards community wealth building.

3.3.2 Key examples:

1. Community land trust: The LDN.CASH governance structure simplifies adopting materials based on members skills and experience levels, to support community land trust goals, including community-owned and -governed digital infrastructure and interactive systems.
2. Housing co-operative: In case of a new Housing Co-operative's community-building experience, despite lacking property, solidarity mapping was highlighted as particularly useful for early and exploratory stages of co-op development.
3. Broader community interactions: Co-ops Connect/CoLocal, an online platform in early stages of development, for SE London co-op members to exchange jobs, assistance, and connections, fostering knowledge-sharing and relationship-building within the community.

3.3.3 Reflection and iteration: Flexibility proved to be a defining feature of the final stage, as participants adapted the solidarity mapping approach to reflect their unique contexts and priorities. By prioritizing equity and community-led values, the process reinforced the need for tools and methods that could evolve alongside emerging challenges and opportunities. Participants emphasized the importance of maintaining a balance between structured guidance—such as clearly defined principles and governance frameworks—and the adaptability required to address diverse and often unpredictable scenarios.

3.4 Data analysis

In this study, we adopted Thematic analysis (TA) [17] as the primary framework to explore opportunities for subverting conventional design method. Through the analysis of diverse data from workshops and participant feedback, we identified key themes such as community-led participation, ownership of technology, and flexibility. These themes revealed how solidarity mapping could serve as a subversion of conventional design methods, both in terms of form and content. We used it to refine our understanding of these subversive possibilities through TA's iterative process, integrating both inductive and deductive approaches to deepen our analysis. In this way we were able to ground our work within a

feminist and co-operative framework, ensuring that our findings were deeply rooted in solidarity-based theoretical and practice-led research. This approach allowed us to reflect the aspirations and perspectives of grassroots participants, providing a meaningful connection between their lived experiences and the design process. In the following discussion, we explore the broader implications of these insights, examining how subversion may transform both the form and content of design methodologies.

4 DISCUSSION

To ground our discussion, we revisit the key findings from our case study. Through the subversion of example mapping, we observed how prioritizing co-operative principles, mutual aid, and feminist goals can transform traditional HCI design methods. These sessions demonstrated that by centring the voices of small-scale co-operatives and community land trusts, it is possible to reshape design processes that typically privilege efficiency and productivity into ones that prioritize inclusivity and shared community goals. This subversion not only highlighted the limitations of existing design paradigms but also opened pathways for participatory planning rooted in economic justice. Furthermore, this study demonstrates how the cyclical and flexible processes of participatory design can subvert both the form and content of traditional design methods to better align with co-operative principles of equity, ethics, and accessibility. By combining iterative reflection and adaptation, each stage of our study pointed to opportunities to challenge conventional practices and reimagine them as instruments of change.

4.1 Subversion of form and content

Building on these findings, we conceptualize subversion as an instrument of change within HCI. Subversion, as it relates to our work, operates on two levels: form and content, as outlined by Amos Vogel [ref]. Subversion of form involves altering the structure and processes of a design method to disrupt existing power dynamics, while subversion of content addresses the ideological underpinnings and intended outcomes of those methods. In our adaptation of example mapping, subversion of form was evident in the way we restructured participant roles and session dynamics to centre voices often marginalized in traditional design discussions. Subversion of content, on the other hand, emerged through a focus on co-operative goals and the dismantling of values aligned with market-driven priorities. To contextualize this subversion further, we draw on the work of scholars such as Light [55], DiSalvo [27], and Dombrowski [29], who emphasize the political and ethical dimensions of design methods. By modifying example mapping to foreground feminist and co-operative principles, we align with critical HCI scholarship that seeks to challenge entrenched hierarchies and reimagine technology as a tool for empowerment rather than control. Additionally, the broader implications of this work resonate with Fedosov's [35] emphasis on participatory design's potential for fostering solidarity and shared agency.

Analysing our case study through the lens of subversion, we begin with a subversion of form. Example mapping's conventional structure inherently reinforces power asymmetries by privileging participants who are skilled in the language of design and technical

decision-making. By intentionally diversifying the roles and contributions of participants in our sessions, we changed this dynamic. For instance, we reframed discussions to prioritize community-defined goals over predefined business objectives, ensuring that co-operative members and CLT representatives could articulate their needs and aspirations without deferring to more dominant voices. Moving to the subversion of content, our adaptation of example mapping shifted the focus from efficiency and productivity—core tenets of the original method—to inclusivity and mutual aid. This was achieved by reframing user stories and examples to reflect co-operative values, such as shared ownership and economic justice. In doing so, we not only challenged the ideological assumptions underpinning the method but also demonstrated how these methods could be reoriented to support community wealth-building initiatives. These findings underscore the transformative potential of subversion as a HCI methodological strategy.

4.2 Implications for HCI

The implications of this work extend to critical HCI discourse, particularly in its engagement with social justice and participatory design methods. Particularly for critical HCI, scholars such as Light [54], Dombrowski [30], Subasi et al [60], Erete et al [33], Ekbias and Nardi [32], Fox et al [37], Chordia et al [19], Khovanskaya et al [50], have all called for a reimagining of design practices that prioritize ethical and equitable outcomes. Our findings build on this by illustrating how subversion can serve as a practical mechanism for aligning design methods with these goals. Furthermore, by situating our work within the context of community wealth building, we address a gap in HCI scholarship identified by the researchers who advocate for strategies that explicitly tackle economic inequality. Critically, our approach also engages with debates surrounding the limitations of participation and co-design [7, 11, 12, 15]. While participatory methods are often heralded as inherently democratic, scholars like Udoewa [62] and Fuchs [39, 40] have highlighted their potential to reproduce existing power imbalances if not carefully managed. By explicitly including principles of mutual aid and co-operation into the process, our adaptation of example mapping addresses these critiques, offering a novel dimension for participatory design within HCI.

4.3 Challenges and limitations

Subverting design methods' content and form is not without obstacles. Institutional and commercial interests may resist such approach, deeming it inefficient or impractical. Similarly, flexibility in modifying design methods is vital for supporting grassroots communities but is often dismissed by those invested in intellectual property protections, as it challenges traditional ownership models. Limited resources, time constraints, and the negative connotation of "subversion" further hinder its recognition as a productive tool for addressing economic and social inequalities. Despite these limitations, we find the subversion of design methods to be highly effective approach. Its capacity to promote community-driven, solidarity-based approaches makes it an invaluable tool for fostering equity and inclusion in the design process.

5 CONCLUSION AND FUTURE WORK

Through this study we explore the potential of subverting a conventional HCI design method to stand against economic inequality. By modifying example mapping, we show how stirring a collaborative design method towards co-operative principles may undo entrenched hierarchies in design processes. The findings demonstrate the importance of flexibility, inclusivity, and mutual aid in fostering community-driven solutions. Furthermore, this work contributes to the HCI by aligning design practices with social justice goals and addressing economic inequality through participatory design. Ultimately, we advocate for more research in reimagining technology as a catalyst for collective empowerment and systemic change. As this work grows towards developing a cohesive methodological framework the emphasis will be on consolidating these findings into actionable recommendations for policymakers, industry and academia, aiming to shape future technology development that prioritizes community ownership, fairness, and economic justice.

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Figure 1: Co-op Hackathon workshop, image courtesy of Co-operatives UK's Co-op Hackathon / Suzi Corker Photography



Figure 2: A3 color plates, initial Example mapping subversion developed earlier to include emancipation, used as a starting point for Co-op Hackathon workshop

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A APPENDIX 1

Co-op Hackathon: Introducing example mapping
The Sandbox sessions: Looking for subversion
Community land trust away day: Solidarity mapping

B APPENDIX 2

SOLIDARITY MAPPING

The Solidarity mapping method is a subversion of the conventional example mapping method, reimagined to support community



Figure 3: Workshop materials



Figure 4: Final subversion of content and form from example to solidarity mapping with developed steps to support early exploratory stages of community organizing, shared understanding, based on LDN.CASH structure.



Figure 5: Final subversion of content and form from example to solidarity mapping with developed steps to support early exploratory stages of community organizing, shared understanding, based on LDN.CASH structure.

wealth building, the solidarity economy and the explicit inclusion of co-operative principles and feminist goals. We propose this method may help co-operatives and other community asset-building groups collaboratively explore and define key aspects of their initiatives that translate into interactive systems design requirements while ensuring these contribute to the goals of equality, solidarity and mutual aid.

